

COMPUTER SCIENCE PAPER – I (THEORY)

CLASS – XI (Advanced Level)

Time: 2 hours Maximum Marks: 70

PART – I (20 Marks)

Question 1

i) The binary equivalent of the decimal number 58 is:

(a) 111001 (b) 111010 (c) 111011 (d) 111100 [1]

ii) The output of XOR gate is 1 when:

(a) All inputs are 1 (b) Inputs are equal (c) Inputs are different (d) All inputs are 0 [1]

iii) The complement of the Boolean expression $A + B'C$ is:

(a) $A'B' + C'$ (b) $A'B + C'$ (c) $(A'B) + C$ (d) $(A'B') \cdot C'$ [1]

iv) The decimal equivalent of $(1101.101)_2$ is:

(a) 13.75 (b) 14.625 (c) 13.625 (d) 12.75 [1]

v) Which of the following inheritance types can be implemented using interfaces in Java?

(a) Multiple (b) Multilevel (c) Single (d) Hybrid [1]

vi) State and verify the Absorption Law using a truth table. [2]

vii) Show that $X(X + Y) = X$ is an identity. [1]

viii) Find the dual of the following expression:

$(A + 0)(B + 1) = A$ [2]

Question 2

i) The function below is incomplete. Fill in the missing parts ?1?, ?2?, ?3?.

```
int countVowels(String s) {  
    int count = 0;  
    int i = ?1?;  
    while(?2?) {  
        char ch = Character.toLowerCase(s.charAt(i));  
        if("aeiou".indexOf(ch) != -1)  
            count++;  
        ?3?;  
    }  
    return count;  
}
```

(a) Expression for ?1? [1]

(b) Expression for ?2? [1]

(c) Expression for ?3? [1]

ii) Given: `int arr[][] = { {2,4,6}, {1,3,5}, {7,9,11}, {13,15,17} };`

(a) What is the value of `arr[3][1]`? [1]

(b) What is the value of `arr[0].length`? [1]

iii) Write a Java statement to initialize a 2D array of 3 rows and 4 columns with all elements as 0. [1]

iv) Give the output of the following: [3]

(a) `System.out.println("HELLO".substring(1,4));`

(b) `System.out.println(Math.pow(2,3) + Math.sqrt(16));`

(c) `System.out.println("Data".compareToIgnoreCase("data"));`

v) Observe the class definition:

```
class Car {  
    String model;  
    double price;  
    void display() {  
        System.out.println(model + " : " + price);  
    }  
}
```

`Car c = new Car();`

(a) Identify one instance variable. [1]

(b) What is the object name of the class? [1]

PART – II (50 Marks)

Answer six questions: two from Section A, two from Section B, and two from Section C.

SECTION – A

Question 3

(i) Prove using algebraic method: $(p \rightarrow q) \equiv (\neg p \vee q)$ [2]

(ii) Convert $(A3F)_{16}$ to decimal. [1]

(iii) Convert $(127.45)_8$ to binary. [2]

(iv) Convert $(10011101101)_2$ to hexadecimal. [1]

(v) Prove: $A + A'B = A + B$ [2]

(vi) Write the inverse, converse, and contrapositive of: "If a number is even, then it is divisible by 2." [3]

Question 4

(a) Find the complement using De Morgan's Law: $(A + B + C') \cdot (A' + B' + C)$ [2]

(b) Simplify using Boolean laws: $(A + B)(A + B')$ [2]

(c) Prove by truth table that $(p \oplus q) \equiv (\neg p \Leftrightarrow q)$ [2]

(d) Convert to Java statement: $z = (a^2 + b^2 - c^2)/(2ab)$ [2]

(e) Differentiate between static and instance methods. [2]

Question 5

- (a) Convert $(45.625)_{10}$ to binary. [2]
- (b) Subtract $(11011)_2 - (10101)_2$ using 2's complement. [2]
- (c) Verify by truth table: $A(B + C) = AB + AC$ and name the law. [2]
- (d) Given a NOR gate,
 - i) What is its output when $A = 1, B = 0$? [1]
 - ii) For what inputs is the output 1? [1]
- (e) Simplify: $F = (A + B')(A' + C)$ and find F' . [2]

SECTION – B

Question 6

Define a class BankAccount with:

- Data members: accNo, name, balance
- Member methods:
 - void input() – to accept account details
 - void deposit(double amt) – to add amount
 - void withdraw(double amt) – deduct amount if sufficient balance else show “Insufficient funds”
 - void display() – to show updated details

Write the complete class with main() to demonstrate all functionalities. [10]

Question 7

Define a class MatrixOps to perform matrix operations:

- Accept two 2×2 matrices
- Compute their sum and product
- Display all matrices neatly

Write full class definition with methods and the main function. [10]

Question 8

Write a Java program to input 10 strings and display them in reverse alphabetical order.

Also, count and print how many strings begin with a vowel. [10]

SECTION – C

Question 9

- (a) Determine by truth table whether $(p \wedge q) \Rightarrow (p \vee q)$ is a tautology. [2]

(b) Let

p = “The system is on.”

q = “The user can log in.”

Construct compound propositions for:

- i) $\neg p \vee q$ ii) $p \leftrightarrow q$ iii) $\neg q \Rightarrow \neg p$ [3]

Question 10

- (a) Define number base conversion and state any two uses in computer systems. [2]
(b) Draw the logic circuit for the expression $(A \oplus B)' + C$. [3]

Question 11

A courier company selects delivery staff using the following logic:

- A person is qualified (Q) and has a valid license (L)

OR

- They are internally trained (T) and have over 1 year of experience (E)

Input: Q – Qualified, L – License, T – Trained, E – Experienced

Output: S – Selected (1 = Yes, 0 = No)

Tasks:

1. Construct the truth table for all possible inputs and output.
2. Derive the expression for each true output.
3. Simplify the final Boolean expression for S. [5]